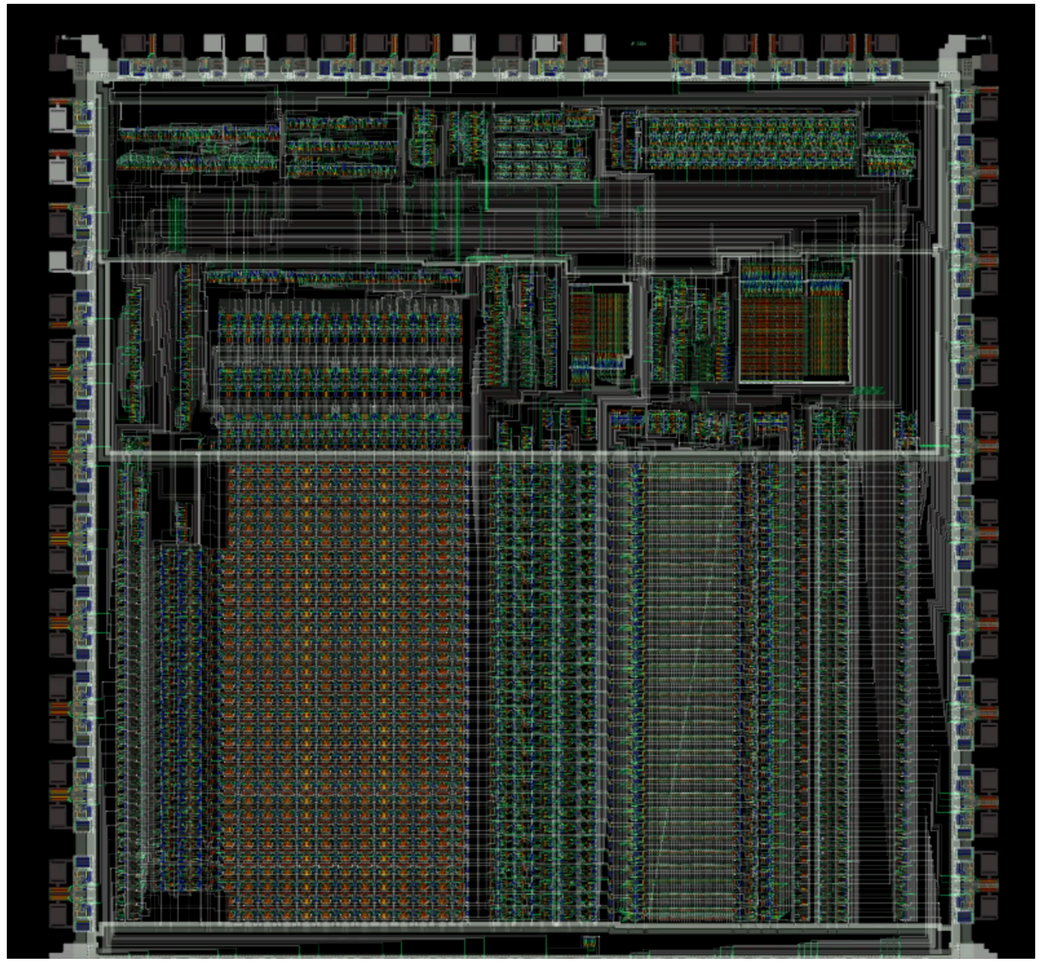


Basics of Logic Design

Alex Breuer

Overview and Schedule

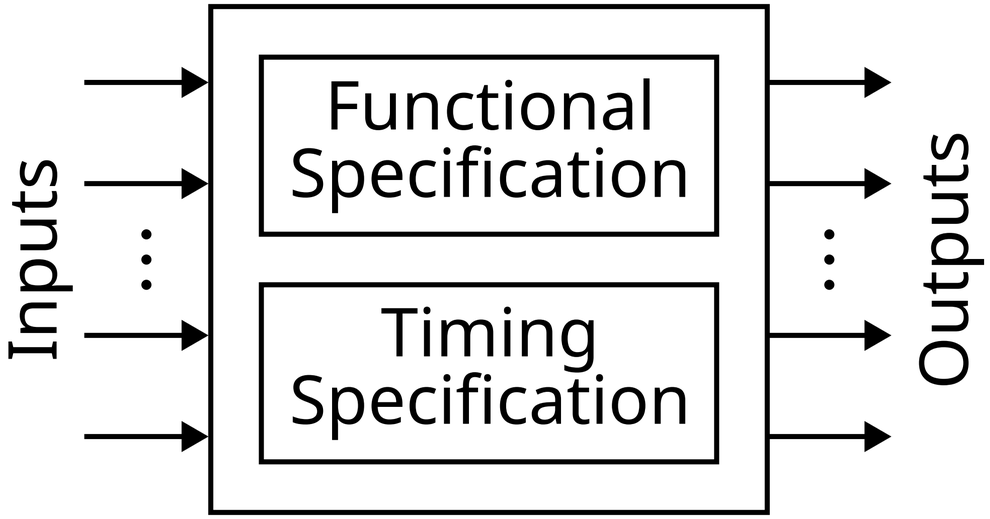
- Hardware components are typically designed using an “abstract” representation in a hardware description language
- Understanding such abstract representations requires some background knowledge
- We’ll get started by getting familiar with these basics
- Once accomplished we are ready to dive deeper into logic design and understand the structure of a processor’s crucial components, e.g., the Arithmetic Logic Unit (ALU)



Source: http://www.visual6502.org/sim/varm/armgl.html

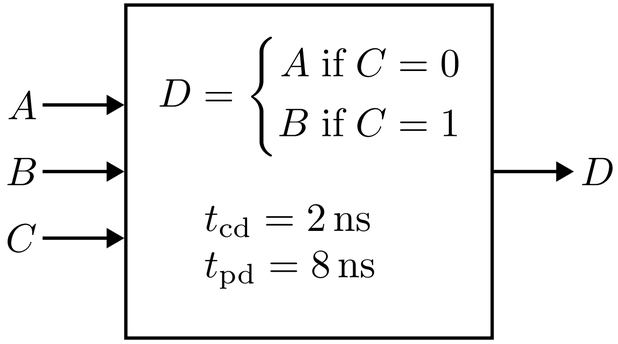
Combinational Logic Circuit

- Digital inputs (at least one)
- Digital outputs (at least one)
- Functional specification: logic values of the outputs for each combination of input values
- Timing specification: delay(s) until outputs are computed from inputs
- Combinational logic blocks contain no memory. Their functional behavior is completely specified by the output values for every combination of input values.



Truth Tables

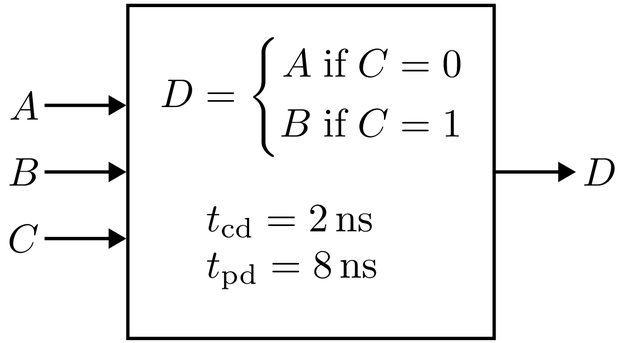
- Truth table:
  - Explicitly lists the output values for every combination of input values: 2^n rows for n binary inputs
  - Alternative notation for functional specification
- Example:
  - Inputs: A, B, C
  - Output: D
  - Functional specification: $D = \begin{cases} A \text{ if } C = 0 \\ B \text{ if } C = 1 \end{cases}$
  - Timing specification: $t_{cd} = 2\,\text{ns}$, $t_{pd} = 8\,\text{ns}$



A	B	C	D
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

Boolean Expressions

- A Boolean expression combines variables and constants with values in $\{0, 1\}$ using logical operators
- Operators:
  - OR (Boolean addition): $A + B$ is 1 if and only if at least one input is 1
  - AND (Boolean multiplication): $A \cdot B$, also written AB, is 1 if and only if both inputs are 1
  - NOT (negation): $\bar{A}$ is 1 if and only if $A = 0$
- Operator precedence, highest to lowest: NOT, AND, OR. Parentheses override this order.



$$D = \bar{C}A + CB$$

Boolean Algebra

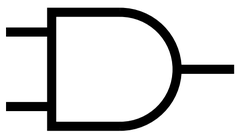
Name	Axiom	Dual
Boolean domain	$A = 0 \text{ if } A \neq 1$	$A = 1 \text{ if } A \neq 0$
NOT	$\bar{0} = 1$	$\bar{1} = 0$
OR/AND	$0 + 0 = 0$	$1 \cdot 1 = 1$
OR/AND	$1 + 1 = 1$	$0 \cdot 0 = 0$
OR/AND	$0 + 1 = 1 + 0 = 1$	$0 \cdot 1 = 1 \cdot 0 = 0$

Name	OR Form	AND Form
Identity	$A + 0 = A$	$A \cdot 1 = A$
Null Element	$A + 1 = 1$	$A \cdot 0 = 0$
Idempotency	$A + A = A$	$A \cdot A = A$
Inverse	$A + \bar{A} = 1$	$A \cdot \bar{A} = 0$
Double Complement	$\bar{\bar{A}} = A$	
Commutativity	$A + B = B + A$	$A \cdot B = B \cdot A$
Associativity	$A + (B + C) = (A + B) + C$	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$
Distributivity	$A + (B \cdot C)$ $= (A + B) \cdot (A + C)$	$A \cdot (B + C)$ $= (A \cdot B) + (A \cdot C)$
DeMorgan	$\overline{A + B} = \bar{A} \cdot \bar{B}$	$\overline{A \cdot B} = \bar{A} + \bar{B}$

Gates

- Gate: device which implements a basic logic function
- Inversion *bubble* inverts input / output value
- Any logic function can be constructed from AND gates, OR gates and inverters

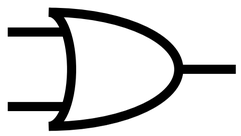
AND



$C = A \cdot B$

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

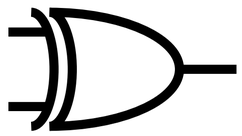
OR



$C = A + B$

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

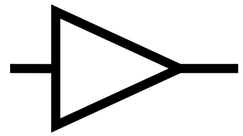
XOR



$C = A \oplus B$

A	B	C
0	0	0
0	1	1
1	0	1
1	1	0

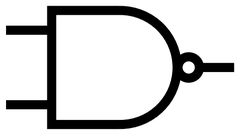
Buffer



$B = A$

A	B
0	0
1	1

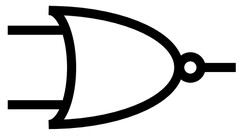
NAND



$C = \overline{A \cdot B}$

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

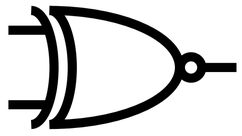
NOR



$C = \overline{A + B}$

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

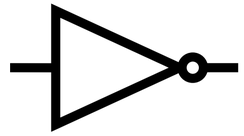
XNOR



$C = \overline{A \oplus B}$

A	B	C
0	0	1
0	1	0
1	0	0
1	1	1

NOT



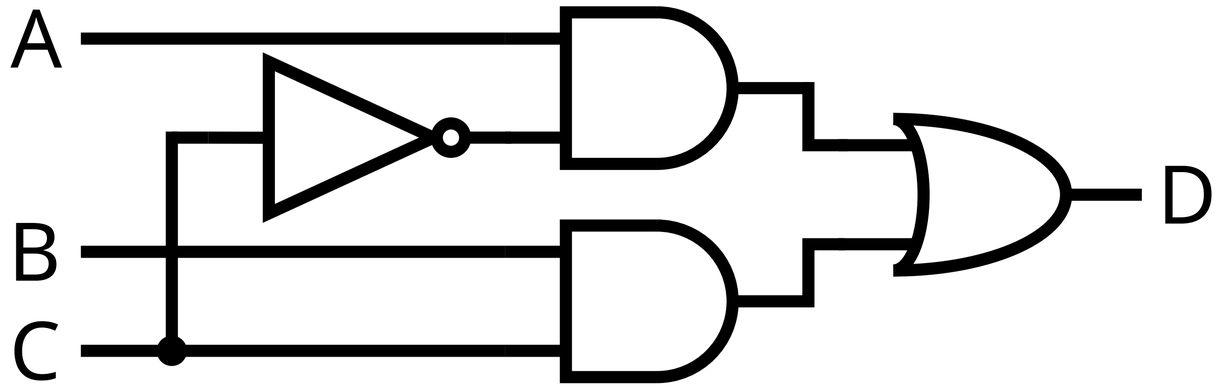
$B = \bar{A}$

A	B
0	1
1	0

Schematics

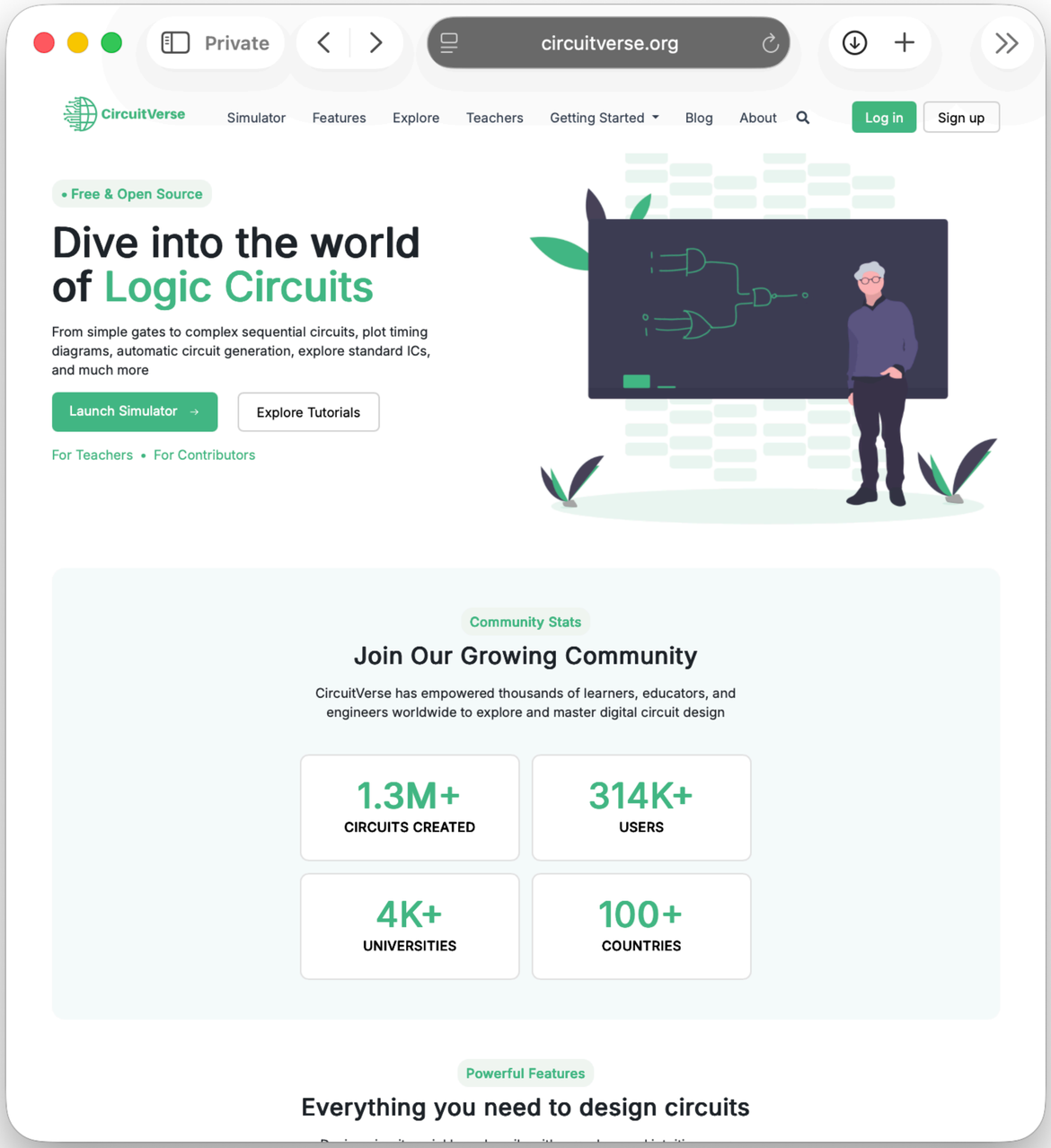
- T junction: wires are connected
- Wires cross without a dot: no connection between wires
- Wires cross with a dot: wires are connected
- Good practices:
  - Use left/top for inputs
  - Use right/bottom for outputs
  - Use dots at T junctions as well
  - Use straight wires if possible

$$D = \bar{C}A + CB$$



Interactive Tool

- “Inspired by Logisim, CircuitVerse is an open-source educational tool for designing and visualizing digital logic circuit simulations in the cloud.”
- Available at: https://circuitverse.org/
- We’ll use CircuitVerse to get started
- Later: switch to hardware description language



Universal Gates

- Set of gates is universal if any Boolean function can be constructed from it, i.e. set {AND, OR, NOT} is universal
- {NAND} is universal:

$$\overline{A \cdot A} = \bar{A}$$

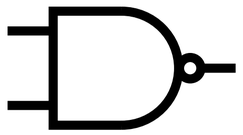
$$\begin{aligned}\overline{\overline{A \cdot B}} &= \overline{\bar{A} + \bar{B}} \\ &= \bar{\bar{A}} \cdot \bar{\bar{B}} = A \cdot B\end{aligned}$$

$$\begin{aligned}\overline{\bar{A} \cdot \bar{B}} &= \bar{\bar{A}} + \bar{\bar{B}} \\ &= A + B\end{aligned}$$

- {NOR} is universal: exercise

Name	OR Form	AND Form
Identity	$A + 0 = A$	$A \cdot 1 = A$
Null Element	$A + 1 = 1$	$A \cdot 0 = 0$
Idempotency	$A + A = A$	$A \cdot A = A$
Inverse	$A + \bar{A} = 1$	$A \cdot \bar{A} = 0$
Double Complement	$\bar{\bar{A}} = A$	
Commutativity	$A + B = B + A$	$A \cdot B = B \cdot A$
Associativity	$A + (B + C) = (A + B) + C$	$A \cdot (B \cdot C) = (A \cdot B) \cdot C$
Distributivity	$A + (B \cdot C)$ $= (A + B) \cdot (A + C)$	$A \cdot (B + C)$ $= (A \cdot B) + (A \cdot C)$
DeMorgan	$\overline{A + B} = \bar{A} \cdot \bar{B}$	$\overline{A \cdot B} = \bar{A} + \bar{B}$

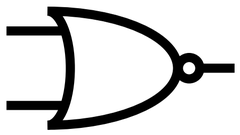
NAND



$C = \overline{A \cdot B}$

A	B	C
0	0	1
0	1	1
1	0	1
1	1	0

NOR



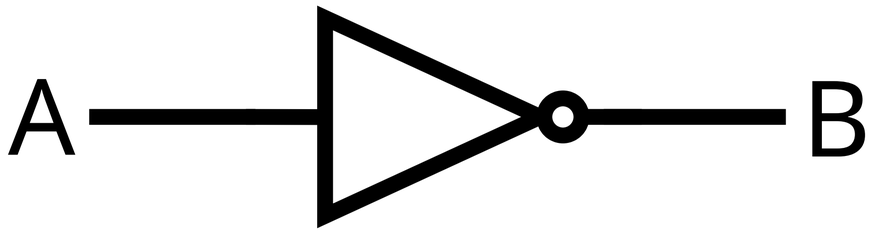
$C = \overline{A + B}$

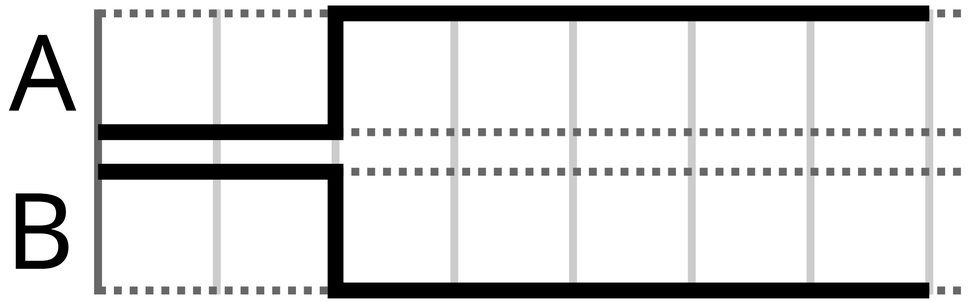
A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

Combinational Timing

Timing Diagram

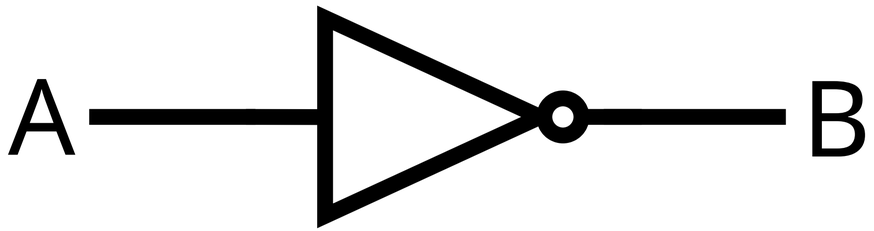
- Timing diagrams / waveforms used to visualize the behavior of circuits over time
- Binary abstraction: Idealized two-level waveforms with instantaneous transitions; LOW represents 0 and HIGH represents 1
- We'll use waveforms extensively when running simulations of our circuits

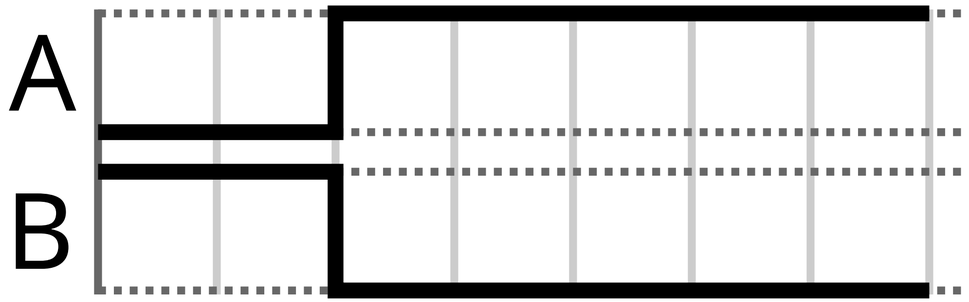




Motivation: Delays

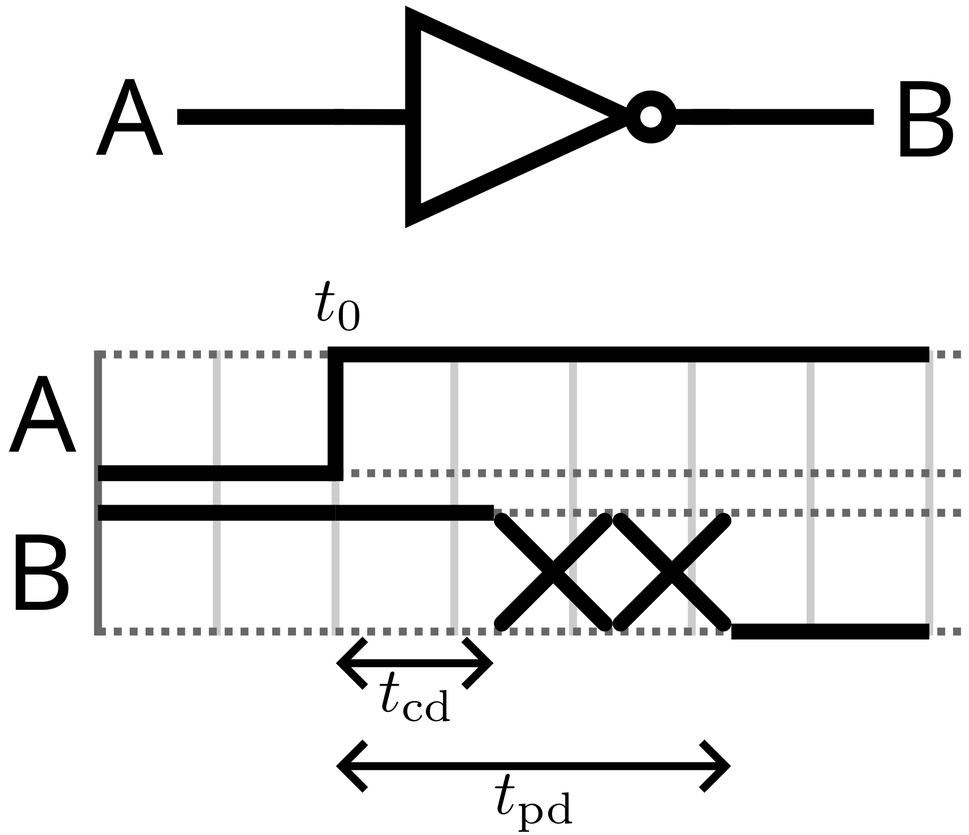
- Example on the right: Instantaneous response of signal B when A changes from LOW to HIGH
- Reality: It takes time for an output to respond to changes of an input; circuits are subject to delays!





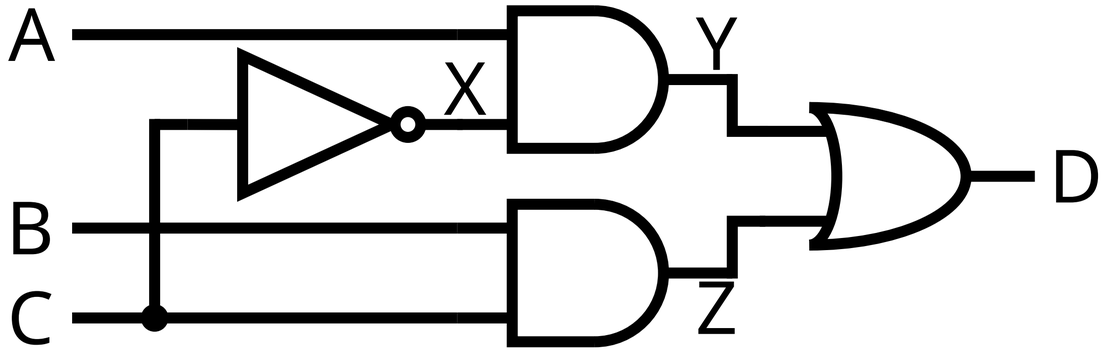
Delays

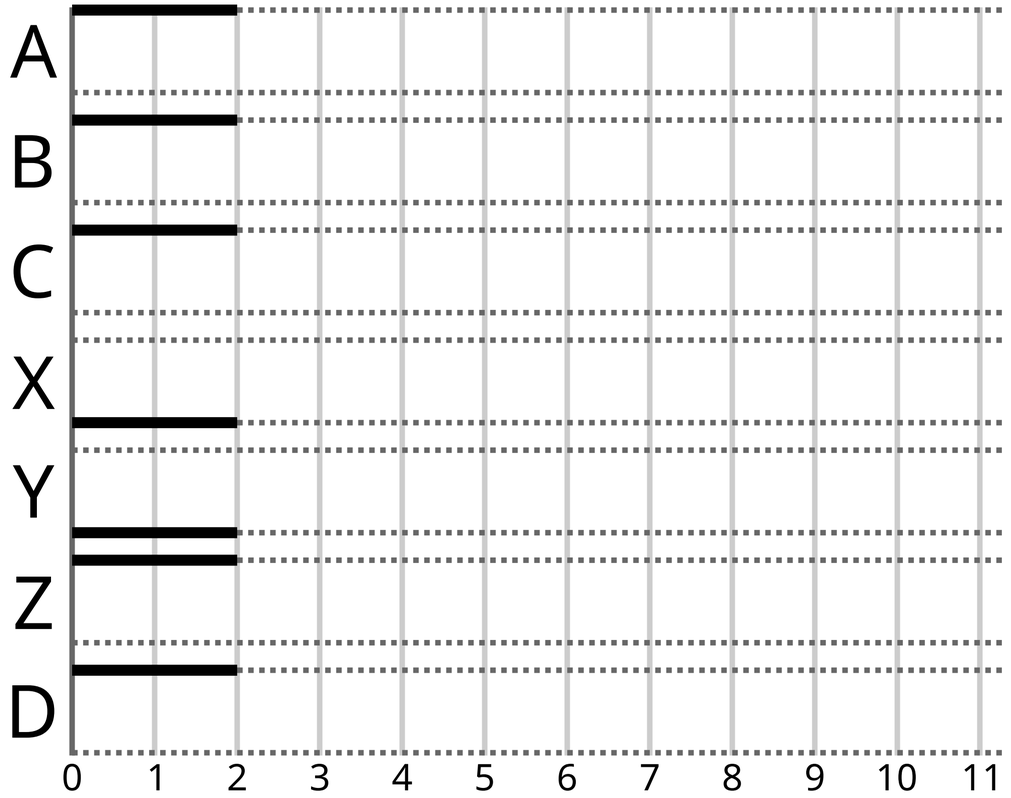
- Combinational circuit characterized by contamination delay t_{cd} and propagation delay t_{pd}
- Contamination delay t_{cd}: If the outputs are initially stable and valid, they retain their old values for at least t_{cd} after an input changes at time t_0.
- Propagation delay t_{pd}: If the inputs become valid at time t_0 and remain stable, all outputs are guaranteed to have settled to their correct values by $t_0 + t_{pd}$.



Example

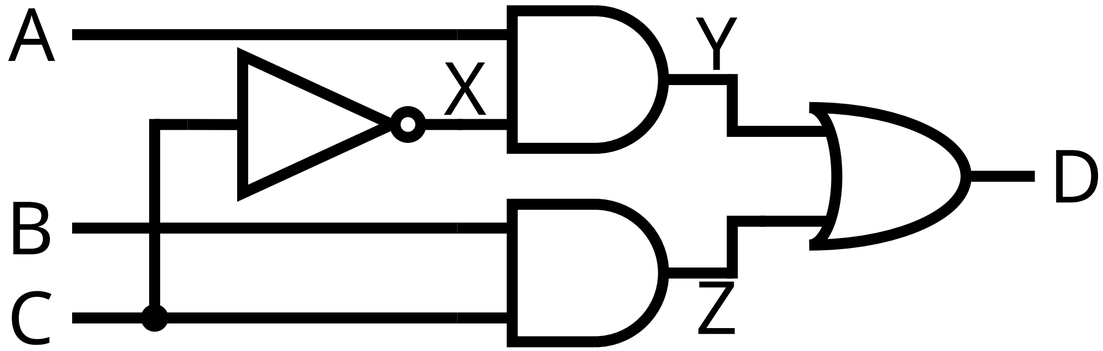
- Assume the circuit on the right:
  - Three inputs: A, B and C
  - One output: D
  - Internal signals: X, Y, Z
- The gates have the following delays:
  - Inverter: $t_{\text{cd}} = 1\ \text{ns}, t_{\text{pd}} = 2\ \text{ns}$
  - AND/OR: $t_{\text{cd}} = 1\ \text{ns}, t_{\text{pd}} = 3\ \text{ns}$
- Delays of wires: Ignored, i.e. infinite speed
- Question: What are t_{cd} and t_{pd} of the entire circuit?

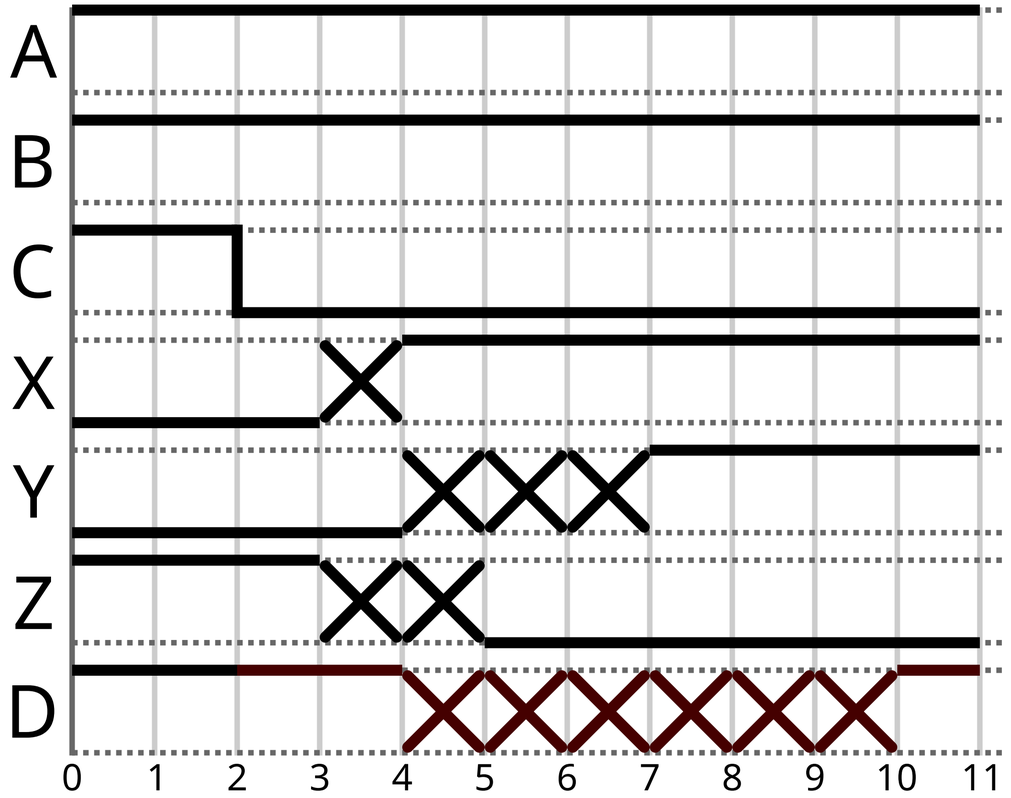




Timing Analysis

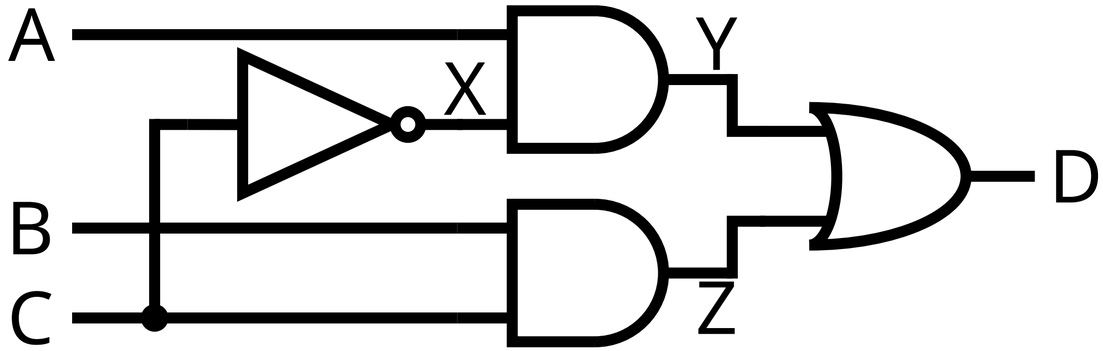
- Assume the circuit on the right:
  - Three inputs: A, B and C
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  - Internal signals: X, Y, Z
- The gates have the following delays:
  - Inverter: $t_{\text{cd}} = 1\ \text{ns}, t_{\text{pd}} = 2\ \text{ns}$
  - AND/OR: $t_{\text{cd}} = 1\ \text{ns}, t_{\text{pd}} = 3\ \text{ns}$
- Delays of wires: Ignored, i.e. infinite speed
- Let's first have a detailed look at what happens if all inputs are HIGH and C changes to LOW

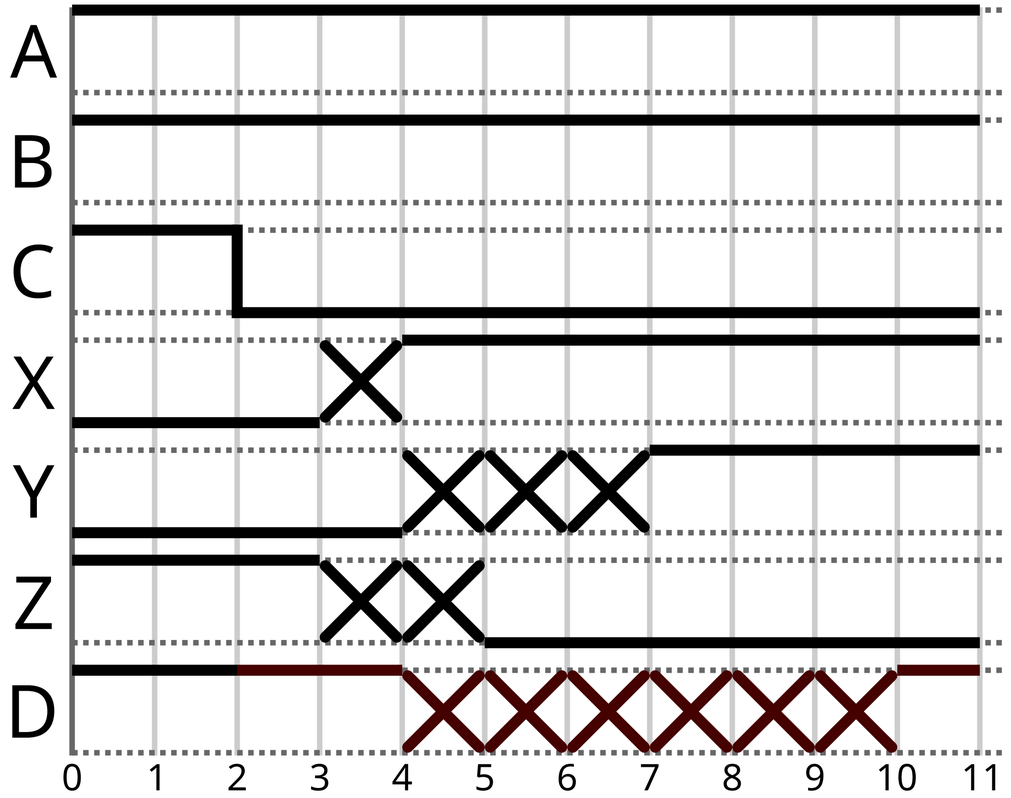




Glitches

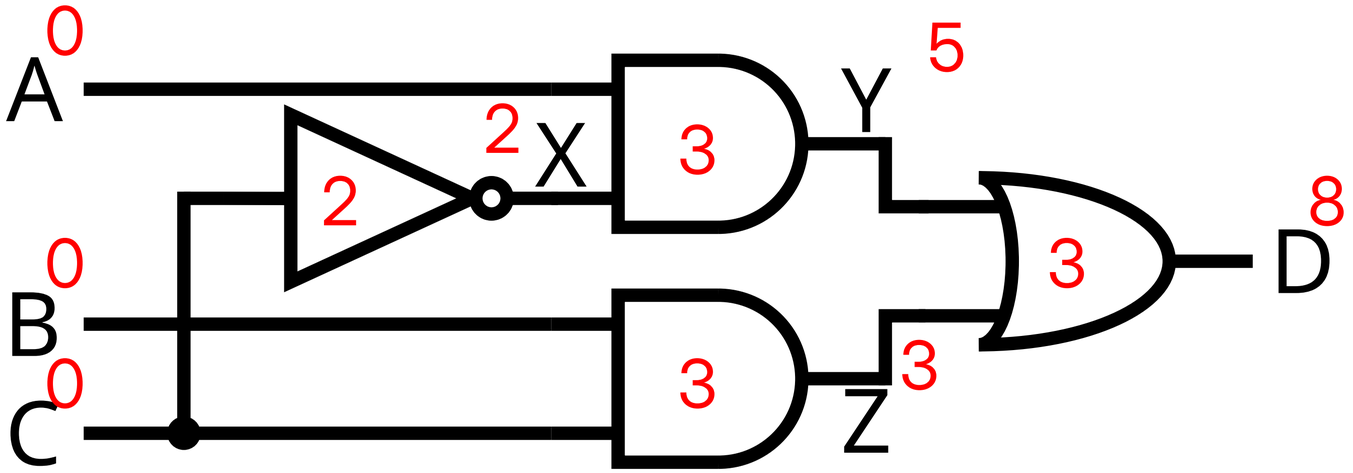
- A glitch is an unwanted transient change in a signal. Unequal propagation delays along different paths can cause glitches
- For a change of input C (right example) this is possible in the range from 4 ns - 10 ns
- Example:
  - Assume that the internal signal Z settles to LOW at 3 ns and Y stays LOW until 7 ns where it changes to HIGH
  - D must be LOW by 6 ns and cannot begin rising before 8 ns, so it remains LOW for at least 2 ns.
  - But: $D = \bar{C}A + CB = 1 \cdot 1 + 0 \cdot 1 = 1$
- Bottom line: It's crucial to be aware of delays and corresponding unspecified behavior if delays are not accounted for





Circuit-level Delays

- Contamination delay t_{cd} of a circuit: shortest cumulative contamination delay along any input-output path
- Propagation delay t_{pd} of a circuit: longest cumulative delay along any input-output path
- Example: Let's derive the propagation delay of the circuit on the right
- Critical path: slowest path from any input to any output
- When simply talking about a "delay" in future we typically mean "propagation delay"



Summary and Outlook

- Functional specification of a circuit might be given in different ways:
  - Truth tables
  - Boolean equations
  - Schematics
- Hardware components are subject to contamination and propagation delays
- Designs must respect timing constraints so that signals are used when their values are guaranteed valid

- Immediate future:
  - Study some combinational logic blocks, i.e. multiplexers, decoders and adders
  - Use SystemVerilog to express these blocks and simulate them
  - Bring designs to FPGAs and test our own hardware blocks
- Further away:
  - Arithmetic Logic Unit
  - Sequential Logic